8. Electromagnetic waves propagate in the direction to the vector

(1) $\vec{E}$

(2) $\vec{B}$

(3) $\vec{E} \times \vec{B}$

(4) $\vec{B} \times \vec{E}$

9. The current in an LCR circuit is given by $I = 20\sin\left(100\pi t + \frac{\pi}{3}\right)A$. The voltage across the inductance L of 0.1 H at t=0 will be-

(1) 31.4 V

(2) 3.14 V

(3) 157 V

(4) 314 V

10. The time required for a 50 Hz alternating current to increase from 0 to 70.7 % of its peak value is –

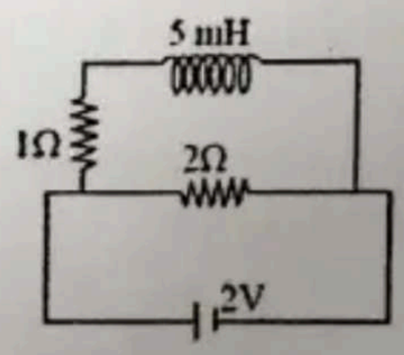
(1) 2.5 ms

(2) 10 ms

(3) 20 ms

(4) 14.14 ms

11. When induced emf in inductor coil is 50% of its maximum value then stored energy in inductor coil in the given circuit will be



(1) 2.5 mJ

(2) 5 mJ

(3) 15 mJ

(4) 20 mJ

12. The peak voltage in a 220 V ac source is

(1) 220 V

(2) about 160 V

(3) about 310 V

(4) 440 V

13. An electromagnetic wave is propagating along +z-axis. Direction of electric and the magnetic field of this wave can be represented by

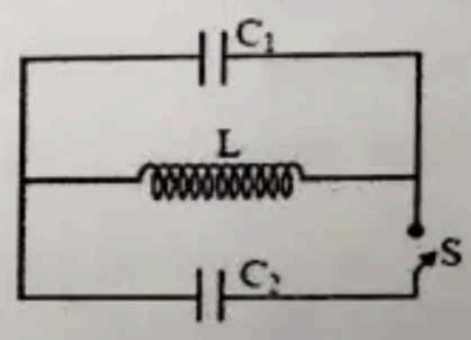
(1) $\vec{E} = E_0\hat{i},\ \vec{B} = B_0\hat{j}$

(2) $\vec{E} = E_0\hat{k},\ \vec{B} = B_0\hat{i}$

(3) $\vec{E} = E_0\hat{j},\ \vec{B} = B_0\hat{i}$

(4) $\vec{E} = E_0\hat{j},\ \vec{B} = B_0\hat{k}$

14. At the moment t = 0, when the charge on the capacitor C_1 is zero, the switch is closed. If I_0 be the current through inductor at t. = 0, then for t > 0 (initially C_2 is uncharged)



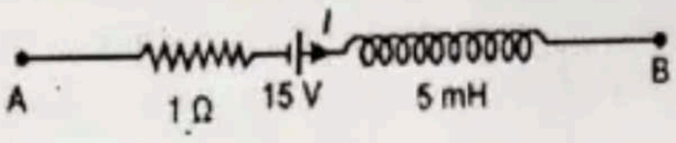
(1) Maximum current through inductor equals $\frac{I_0}{2}$

(2) Maximum current through inductor equals $\frac{C_1 I_0}{C_1+C_2}$

(3) Maximum charge on $C_1 = \frac{C_1 I_0\sqrt{LC_2}}{C_1+C_2}$

(4) Maximum charge on $C_1 = C_1 I_0\sqrt{\frac{L}{C_1+C_2}}$

15. The network shown in the figure is part of a complete circuit. If at a certain instant the current (I) is 5 A and is decreasing at a rate of $10^3 A\ s^{-1}$, then the value of $V_B - V_A$ is



(1) 10 V

(2) 20 V

(3) 15 V

(4) 25 V

16. The resonant frequency of a series AC circuit consisting of a 200μH inductor , 0.0005 μF capacitor and a 10 Ω resistance is

(1) 480KHz

(2) 503 kHz

(3) 406 kHz

(4) 607 kHz

17. The capacitive reactance in an A.C. circuit is

(1) effective resistance due to capacitor

(2) effective wattage

(3) effective voltage

(4) none of the above

18. What is the average power dissipation in an ideal capacitor in AC circuit ?

(1) $2CV^2$

(2) $\frac{CV^2}{2}$

(3) zero

(4) CV^2

19. A telephone wire of length 200 km has a capacitance of 0.014 μF per km. If it carries an AC frequency 5 kHz, what should be the value of an inductor required to be connected in series so that the impedance of the circuit is minimum?

(1) 0.35 mH

(2) 3.5 mH

(3) 2.5 mH

(4) zero

20. In a series LCR circuit the frequency of a 10 V AC voltage source is adjusted in such a fashion that the reactance of the inductor measures 15 Ω and that of the capacitor 11 Ω. If R= 3 Ω , the potential difference across the series combination of L and C will be

(1) 8 V

(2) 10 V

(3) 22 V

(4) 52 V

21. The phase difference between voltage and current in an AC circuit containing a resistor and an inductor in series is φ_1 . When the inductor is replaced by a capacitor, the phase difference is changed to φ_2 . The phase difference when all the three elements are connected in series with the same AC source will be

(1) $\tan^{-1}(\tan\phi_1 - \tan\phi_2)$

(2) $\tan^{-1}(\tan\phi_2 - \tan\phi_1)$

(3) $\cos^{-1}(\cos\phi_1 - \cos\phi_2)$

(4) $\cos^{-1}(\cos\phi_2 - \cos\phi_1)$

22. In a plane electromagnetic wave, the electric field oscillates sinusoidally at a frequency of 2.5×10^{10} Hz and amplitude 480 V m^{-1}. The amplitude of the oscillating magnetic field will be

(1) $1.52 \times 10^{-8}\ Wb\ m^{-2}$

(2) $1.52 \times 10^{-7}\ Wb\ m^{-2}$

(3) $1.6 \times 10^{-6}\ Wb\ m^{-2}$

(4) $1.6 \times 10^{-7}\ Wb\ m^{-2}$

23. The radiation having the least wavelength out of the following option is

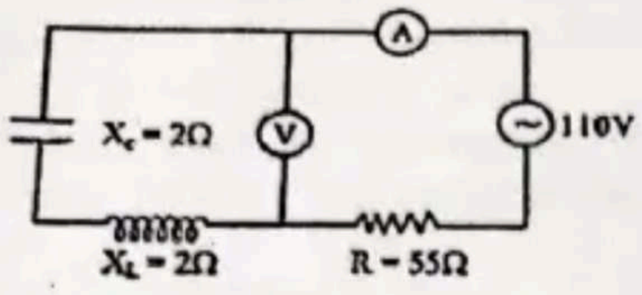
(1) γ - rays

(2) $\beta\ rays$

(3) $\alpha\ rays$

(4) $X\ rays$

24. The reading of the ammeter and voltmeters are (Both the instruments are AC meters and measures RMS value)



(1) 2 A, 110 V

(2) 2A, 0 V

(3) 2A, 55V

(4) 1A, 0 V

25. In a series LCR circuit, the voltage across resistance, capacitance and inductance is 10 V each. If the capacitor is removed, the voltage across the inductance will be

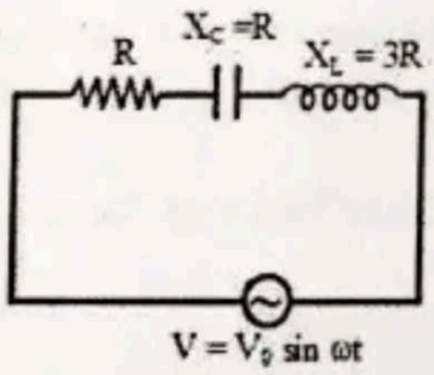
(1) 10 V

(2) $10\sqrt{2}$ V

(3) $\frac{10}{\sqrt{2}}$ V

(4) 20V

26. The power dissipated in the circuit



(1) $\frac{V_0^2}{2R}$

(2) $\frac{V_0^2}{5R}$

(3) $\frac{V_0^2}{\sqrt{5}R}$

(4) $\frac{V_0^2}{10R}$

27. A 30 μF capacitor is connected to a 150 V, 60 Hz AC supply. The RMS value of current in the circuit is

(1) 17 A

(2) 1.7 A

(3) 1.7 mA

(4) 1.7 μA

28. An AC voltage source of variable angular frequency ω and fixed amplitude V_0 is connected in series with a capacitance C and an electric bulb of resistance R (inductance zero). When ω is increased

(1) The bulb glows dimmer

(2) The bulb glows brighter

(3) Total impedance of the circuit is unchanged

(4) Total impedance of the circuit increases

29. A beam of light of intensity I is normally incident on a plate of area A. If plate absorbs fraction η of incident light and reflects rest of it back, then the force exerted by the incident light on the plate is given by [c=speed of light]

(1) $\frac{IA}{c}(1-\eta)$

(2) $\frac{IA}{c}(\eta+1)$

(3) $\frac{IA}{c}(2\eta-1)$

(4) $\frac{IA}{c}(2-\eta)$

30. Average power in the LCR circuit depends upon

(1) Current

(2) Phase difference only

(3) EMF

(4) Current EMF and phase difference

31. In an oscillating LC circuit the maximum charge on the capacitor is Q. The charge on the capacitor when the energy is stored equally between the electric and magnetic field is

(1) $\frac{Q}{2}$

(2) $\frac{Q}{\sqrt{3}}$

(3) $\frac{Q}{\sqrt{2}}$

(4) Q

32. A 20 V AC is applied to a circuit consisting of a resistance and a coil with negligible resistance. If the voltage across the resistance is 12 V, the voltage across the coil is

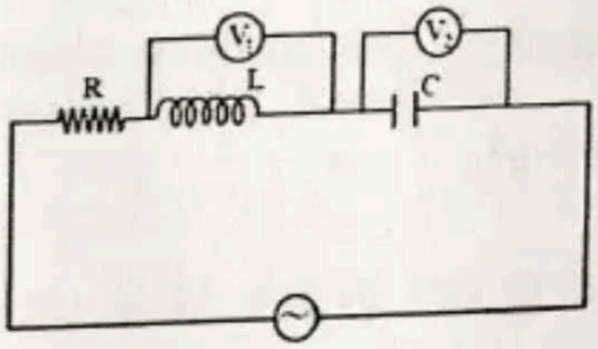
(1) 16 V

(2) 10 V

(3) 8 V

(4) 6 V

33. If resistance R = 100Ω, inductance L $=\frac{2}{\pi}$ H and capacitance C = $\frac{8}{\pi}$ μF are connected in series with an AC source of 200 volt and frequency f as shown in the circuit. If the readings of the hot wire voltmeters V_1 and V_2 are same then which among the following statements is wrong?



(1) f = 125 Hz

(2) f = 250π Hz

(3) current through R is 2A

(4) $V_1 = V_2$ = 1000 volt

34. An electromagnetic wave of frequency ν = 3.0MHz passes from vacuum into a dielectric medium with relative permittivity $\varepsilon_r = 4.0$. Then

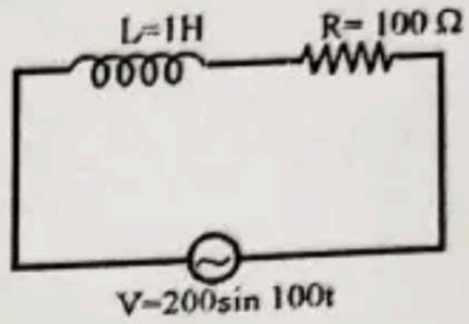
(1) wavelength is doubled and the frequency remains unchanged

(2) wavelength is doubled and frequency becomes half

(3) wavelength is halved and frequency remains unchanged

(4) wavelength and frequency both remain unchanged

35. In the adjacent circuit, the instantaneous current equation is



(1) $2\sin\left(100t-\frac{\pi}{4}\right)$

(2) $\sqrt{2}\sin\left(100t-\frac{\pi}{4}\right)$

(3) $\sqrt{2}\sin\left(200t-\frac{\pi}{4}\right)$

(4) $\sqrt{2}\sin\left(200t+\frac{\pi}{4}\right)$

36. In a series LCR circuit, the voltage drop across the resistance is 8 V, across the inductor is 6 V and across the capacitor is 12 V. Then,

(1) The voltage of the source will be leading the current in the circuit

(2) The voltage drop across each element will be less than the applied voltage

(3) The power factor of the circuit will be 4

(4) None of these

37. Power dissipated in an LCR series circuit connected to an a.c. source of emf ε is :

(1) $\frac{\epsilon^2\sqrt{R^2+\left(L\omega-\frac{1}{C\omega}\right)}}{2R}$

(2) $\frac{\epsilon^2\left[R^2+\left(L\omega-\frac{1}{C\omega}\right)^2\right]}{R}$

(3) $\frac{\epsilon^2 R}{\sqrt{R^2+\left(L\omega-\frac{1}{C\omega}\right)^2}}$

(4) $\frac{\epsilon^2 R}{\left[R^2+\left(L\omega-\frac{1}{C\omega}\right)^2\right]}$

38. For a series L -C –R circuit at resonance, the statement which is not true is

(1) Peak energy stored by a capacitor = Peak energy stored by an inductor

(2) Average power = Apparent power

(3) Wattles current is zero

(4) Power factor is zero

39. In a series resonant LCR circuit, the voltage across R is 100 V and R= 1kΩ with C = 2μF. The resonant frequency ω is 200 rad s^{-1}. At resonance the voltage across L is

(1) 4×10^{-3} V

(2) 2.5×10^{-2} V

(3) 40 V

(4) 250 V

40. An electric motor which is loaded has an effective resistance of 30Ω and an inductive reactance of 40Ω . If the motor is powered by a source with a maximum voltage of 420V , the maximum current is

(1) 6A

(2) 8.4 A

(3) 10A

(4) 12 A

41. A large parallel plate capacitor, whose plates has an area of 1 m^2 and is separated from each other by 1 mm, is being charged at a rate of 25 V s^{-1}. If the dielectric between the plates has the dielectric constant 10, then the displacement current at this instant is

(1) 25 μA

(2) 11 μA

(3) 2.2 μA

(4) 1.1 μA

42. The reason a moving coil galvanometer cannot be used with an alternating current is that

(1) The coil bends easily

(2) The coil heats up too much

(3) Sparks can be produced

(4) The net magnetic field produced is zero

43. In an LCR circuit, the capacitance is made one-fourth, when in resonance. What should be the change in inductance, so that the circuit remains in resonance?

(1) 4 times

(2) $\frac{1}{4}$ times

(3) 8 times

(4) 2 times

44. An ac source of angular frequency ω is fed across a resistor r and a capacitor C in series. The current registered is I. If now the frequency of the source is changed to $\frac{\omega}{3}$ (but maintaining the same voltage), then current in the circuit is found to be halved. Calculate the ratio of reactance to resistance at the original frequency ω.

(1) $\sqrt{\frac{3}{5}}$

(2) $\sqrt{\frac{2}{5}}$

(3) $\sqrt{\frac{1}{5}}$

(4) $\sqrt{\frac{4}{5}}$

45. An A.C. source is connected to a resistive circuit. Which of the following is true?

(1) Current leads ahead of voltage in phase

(2) Current lags behind voltage in phase

(3) Current and voltage are in same phase

(4) Any of the above may be true depending upon the value of resistance